In [ ]:
```
EXPERIMENT 3 – CREATING DATAFRAMES & ANALYZING PROPERTIES
PIN NO.: 241UDS0111
AIM:-
To create Pandas DataFrames using NumPy arrays and CSV files and analyze their properties.
```

In [ ]:
```
## Step 1: Install and Import Libraries
If the libraries are not installed, run `%pip install numpy pandas` in a separate cell first.
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
## Step 2: Create a 2-D Array
Consider three community gardens and their visitor counts over four weeks. Rows represent gardens; columns
represent weeks.
```

In [2]:
```
garden_visitors_241UDS0111 = np.array([
    [142, 168, 193, 207],
    [98, 121, 149, 158],
    [115, 137, 162, 176]
])

garden_visitors_241UDS0111
```

Out[2]:
```
array([[142, 168, 193, 207],
       [ 98, 121, 149, 158],
       [115, 137, 162, 176]])
```

In [3]:
```
garden_visitors_241UDS0111.shape, garden_visitors_241UDS0111.ndim
```

Out[3]:
```
((3, 4), 2)
```

In [ ]:
```
## Step 3: Convert the 2-D Array into a DataFrame
```

In [4]:
```
garden_index_241UDS0111 = ["Riverside", "Hilltop", "Lakeside"]
garden_columns_241UDS0111 = ["Week_1", "Week_2", "Week_3", "Week_4"]

garden_df_241UDS0111 = pd.DataFrame(
    data=garden_visitors_241UDS0111,
    index=garden_index_241UDS0111,
    columns=garden_columns_241UDS0111
)

garden_df_241UDS0111
```

Out[4]:

	Week_1	Week_2	Week_3	Week_4
Riverside	142	168	193	207
Hilltop	98	121	149	158
Lakeside	115	137	162	176

In [ ]:
```
### Add a new column
```

In [5]:
```
garden_df_241UDS0111["Week_5"] = [219, 171, 188]
garden_df_241UDS0111
```

Out[5]:

	Week_1	Week_2	Week_3	Week_4	Week_5
Riverside	142	168	193	207	219
Hilltop	98	121	149	158	171
Lakeside	115	137	162	176	188

In [ ]:
```
### Add a new row
```

In [6]:
```
garden_east_241UDS0111 = pd.DataFrame({
    "Week_1": [87],
    "Week_2": [103],
    "Week_3": [126],
    "Week_4": [139],
    "Week_5": [151]
}, index=["Eastside"])

garden_df_241UDS0111 = pd.concat([garden_df_241UDS0111, garden_east_241UDS0111])
garden_df_241UDS0111
```

Out[6]:

	Week_1	Week_2	Week_3	Week_4	Week_5
Riverside	142	168	193	207	219
Hilltop	98	121	149	158	171
Lakeside	115	137	162	176	188
Eastside	87	103	126	139	151

In [ ]:
```
## Step 4: Creating a DataFrame with Existing CSV/Excel File
Load the provided CSV dataset.
```

In [7]:
```
garden_data_241UDS0111 = pd.read_csv("Experiment_3_241UDS0111_Community_Garden.csv")
garden_data_241UDS0111
```

Out[7]:

	PIN	Garden	Month	Visitors	Harvest_Kg	Volunteer_Hours	Focus_Area
0	241UDS0111	Riverside	January	142	36	18	Vegetables
1	241UDS0111	Riverside	February	168	42	21	Herbs
2	241UDS0111	Riverside	March	193	51	25	Flowers
3	241UDS0111	Hilltop	January	98	27	14	Vegetables
4	241UDS0111	Hilltop	February	121	31	16	Herbs
5	241UDS0111	Hilltop	March	149	39	19	Flowers
6	241UDS0111	Lakeside	January	115	30	15	Vegetables
7	241UDS0111	Lakeside	February	137	35	17	Herbs
8	241UDS0111	Lakeside	March	162	44	22	Flowers
9	241UDS0111	Eastside	January	87	22	11	Vegetables

In [ ]:
```
## Step 5: Analyze the Important Properties of the DataFrame
```

In [8]:
```
garden_data_241UDS0111.head()
```

Out[8]:

	PIN	Garden	Month	Visitors	Harvest_Kg	Volunteer_Hours	Focus_Area
0	241UDS0111	Riverside	January	142	36	18	Vegetables
1	241UDS0111	Riverside	February	168	42	21	Herbs
2	241UDS0111	Riverside	March	193	51	25	Flowers
3	241UDS0111	Hilltop	January	98	27	14	Vegetables
4	241UDS0111	Hilltop	February	121	31	16	Herbs

In [9]:
```
garden_data_241UDS0111.shape
```

Out[9]:
```
(10, 7)
```

In [10]:
```
garden_data_241UDS0111.index
```

Out[10]:
```
RangeIndex(start=0, stop=10, step=1)
```

In [11]:
```
garden_data_241UDS0111.dtypes
```

Out[11]:
```
PIN               object
Garden            object
Month             object
Visitors           int64
Harvest_Kg         int64
Volunteer_Hours    int64
Focus_Area        object
dtype: object
```

In [12]:
```
garden_data_241UDS0111.columns
```

Out[12]:
```
Index(['PIN', 'Garden', 'Month', 'Visitors', 'Harvest_Kg', 'Volunteer_Hours',
       'Focus_Area'],
      dtype='object')
```

In [ ]:
```
## Step 6: Information About the DataFrame
```

In [13]:
```
garden_data_241UDS0111.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
 #   Column           Non-Null Count  Dtype 
---  ------           --------------  ----- 
 0   PIN              10 non-null     object
 1   Garden           10 non-null     object
 2   Month            10 non-null     object
 3   Visitors         10 non-null     int64 
 4   Harvest_Kg       10 non-null     int64 
 5   Volunteer_Hours  10 non-null     int64 
 6   Focus_Area       10 non-null     object
dtypes: int64(3), object(4)
memory usage: 692.0+ bytes
```

In [ ]:
```
## Step 7: Descriptive Statistics
```

In [14]:
```
garden_data_241UDS0111.describe()
```

Out[14]:

	Visitors	Harvest_Kg	Volunteer_Hours
count	10.000000	10.000000	10.000000
mean	137.200000	35.700000	17.800000
std	32.747519	8.641631	4.131182
min	87.000000	22.000000	11.000000
25%	116.500000	30.250000	15.250000
50%	139.500000	35.500000	17.500000
75%	158.750000	41.250000	20.500000
max	193.000000	51.000000	25.000000

In [ ]:
```
## Step 8: Transpose Rows and Columns
```

In [15]:
```
garden_data_241UDS0111.head(10).transpose()
```

Out[15]: Columns 1–8 of 11

	0	1	2	3	4	5	6
PIN	241UDS0111	241UDS0111	241UDS0111	241UDS0111	241UDS0111	241UDS0111	241UDS0111
Garden	Riverside	Riverside	Riverside	Hilltop	Hilltop	Hilltop	Lakeside
Month	January	February	March	January	February	March	January
Visitors	142	168	193	98	121	149	115
Harvest_Kg	36	42	51	27	31	39	30
Volunteer_Hours	18	21	25	14	16	19	15
Focus_Area	Vegetables	Herbs	Flowers	Vegetables	Herbs	Flowers	Vegetables

Columns 9–11 of 11 · first column repeated

	7	8	9
PIN	241UDS0111	241UDS0111	241UDS0111
Garden	Lakeside	Lakeside	Eastside
Month	February	March	January
Visitors	137	162	87
Harvest_Kg	35	44	22
Volunteer_Hours	17	22	11
Focus_Area	Herbs	Flowers	Vegetables

In [ ]:
```
RESULT:-
NumPy arrays and Pandas DataFrames were created, a column and row were added, a CSV dataset was loaded, and
DataFrame properties, information, descriptive statistics, and transpose were examined.
```